MOVIE RATING PREDICTION

INTERNSHIP PROJECT REPORT

A Machine Learning Project Using Python

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Project Title: Movie Rating Prediction

Project Domain: Machine Learning / Data Science

Tools and Technologies:

Python, Google Colab, Pandas, NumPy, Scikit-learn

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ABSTRACT

The Movie Rating Prediction project is a machine learning-based project developed using Python. The main objective of this project is to predict movie ratings by analyzing information available in a movie dataset.

The project uses the IMDb Movies India dataset, which contains information about movies. The dataset is processed using data cleaning techniques to handle missing values and prepare the data for model development.

In this project, the TF-IDF (Term Frequency–Inverse Document Frequency) technique is used to convert textual information into numerical features. Ridge Regression, a machine learning algorithm, is then applied to predict movie ratings.

The model is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 score. The obtained results are used to understand the difference between actual and predicted movie ratings.

This project provides practical experience in data preprocessing, text feature extraction, machine learning model training, prediction, and model evaluation. It also demonstrates how Python and machine learning techniques can be applied to a real-world movie rating prediction problem

1. INTRODUCTION

Movies are an important part of the entertainment industry. People often use movie ratings to understand the general opinion of viewers about a particular movie. These ratings can help users compare movies and make informed viewing choices.

Movie Rating Prediction is a machine learning project that aims to estimate a movie's rating using available movie-related information. In this project, Python and machine learning techniques are used to analyze movie details and predict ratings based on patterns learned from the dataset.

The project uses the IMDb Movies India dataset, which contains information such as movie name, year, duration, genre, rating, director, and actors. These details are processed to create textual features that can be used by a machine learning model.

The TF-IDF (Term Frequency–Inverse Document Frequency) technique is used to transform the combined movie information into numerical features. Ridge Regression is then trained to learn the relationship between these features and the movie ratings.

The model's performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 score. This project provides practical experience in data cleaning, feature extraction, model training, prediction, and evaluation.

Through this internship project, the aim is to understand how machine learning can be applied to a real-world prediction task and to develop practical skills in Python-based data analysis.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

Movie ratings provide useful information about viewers' opinions of films. However, estimating a movie's rating using details such as its genre, director, actors, name, release year, and duration is a challenging prediction task.

The objective of this project is to develop a machine learning model that learns patterns from existing movie information and predicts a numerical rating for a movie. The project uses textual movie features and a regression algorithm to estimate ratings.

The model's predictions are evaluated against actual ratings to understand its performance and limitations.

2.2 Objectives

The main objectives of the Movie Rating Prediction project are:

1. To load and explore the IMDb Movies India dataset.
2. To clean the dataset and handle missing or invalid rating values.
3. To select relevant movie-related features for prediction.
4. To combine movie information into a single text feature.
5. To convert textual information into numerical features using TF-IDF.
6. To train a Ridge Regression model for rating prediction.
7. To evaluate model performance using MAE, RMSE, and R^2 score.
8. To compare actual movie ratings with predicted ratings.
9. To save the trained machine learning model for future use.
10. To gain practical experience in Python, data preprocessing, and machine learning during the internship.

3. TOOLS AND TECHNOLOGIES USED

The Movie Rating Prediction project was developed using Python and its machine learning libraries in the Google Colab environment. The following tools and technologies were used to implement the project.

3.1 Python

Python is the main programming language used in this project. It provides libraries for data processing, feature extraction, machine learning, and model evaluation.

3.2 Google Colab

Google Colab is a cloud-based notebook environment that allows users to write and execute Python code through a web browser. It was used to run the project code, process the dataset, train the model, and display the results.

3.3 Pandas

Pandas was used to load the IMDb Movies India CSV file, inspect the dataset, clean the data, handle missing values, and prepare movie-related features.

3.4 NumPy

NumPy was used for numerical operations, including calculating the Root Mean Squared Error from the mean squared error.

3.5 Scikit-learn

Scikit-learn was used to split the dataset into training and testing sets, convert text into numerical features using TF-IDF, train the Ridge Regression model, and evaluate its performance.

3.6 Matplotlib

Matplotlib was used to create a scatter plot comparing actual movie ratings with predicted ratings.

3.7 Joblib

Joblib was used to save the trained machine learning pipeline as a file named `movie_rating_model.pkl`, which can be reused later.

4. DATASET DESCRIPTION AND DATA PREPROCESSING

4.1 Dataset Description

The dataset used in this project is the IMDb Movies India dataset, stored in CSV format. It contains information about Indian movies, including movie names, release years, duration, genre, ratings, votes, directors, and actors.

The original dataset contains **15,509 rows and 10 columns**. The `Rating` column is used as the target variable that the machine learning model aims to predict.

The dataset includes the following columns:

- **Name:** Name of the movie.
- **Year:** Year of release.
- **Duration:** Duration of the movie.
- **Genre:** Category or type of movie.
- **Rating:** Movie rating used as the prediction target.
- **Votes:** Number of votes received.
- **Director:** Name of the director.
- **Actor 1, Actor 2, Actor 3:** Information about the movie's actors.

4.2 Data Preprocessing

Data preprocessing is an important step in machine learning because raw data may contain missing values, invalid entries, or inconsistent formats. In this project, the following preprocessing steps were performed:

1. **Loading the dataset:** The CSV file was uploaded and read into a Pandas DataFrame.
2. **Standardizing column names:** Column names were converted into lowercase and standardized formats to make them easier to use in Python.
3. **Converting ratings:** The Rating column was converted into numeric values. Entries that could not be converted into valid ratings were treated as missing values.

4. **Removing missing ratings:** Rows without valid ratings were removed because the model requires actual ratings for training.
5. **Handling missing feature values:** Missing values in the selected movie-related feature columns were filled with empty strings.
6. **Creating movie features:** Information from genre, director, actors, movie name, year, and duration was combined into a single text feature called `movie_features`.
7. **Final data preparation:** Rows with empty combined movie features were removed before model training.

After preprocessing, the dataset contained **7,919 rows and 11 columns**, including the newly created `movie_features` column.

4.3 Importance of Data Preprocessing

Data preprocessing helps prepare the dataset for machine learning. It reduces problems caused by missing or invalid ratings and organizes movie information into a format suitable for text feature extraction. The cleaned dataset was then used for training and evaluating the Movie Rating Prediction model.

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5. FEATURE SELECTION AND TF-IDF VECTORIZATION

5.1 Feature Selection

Feature selection is an important step in machine learning because it identifies the information that will be used by the model to make predictions.

In this project, the following movie-related features were selected:

- Genre
- Director
- Actor 1
- Actor 2
- Actor 3
- Movie Name
- Year
- Duration

These features were combined into a single text column named `movie_features`. The target variable was `rating`, which contains the actual movie ratings used to train and evaluate the model.

The `Votes` column was present in the original dataset but was not included in the combined features used by this model.

5.2 TF-IDF Vectorization

TF-IDF stands for Term Frequency–Inverse Document Frequency. It is a text-processing technique used to convert textual information into numerical values that a machine learning algorithm can process.

In this project, TF-IDF was applied to the combined `movie_features` column. It identifies words and word combinations that help represent the textual information of movies.

The model used the following TF-IDF settings:

- **Maximum features:** 10,000
- **N-gram range:** (1, 2), allowing individual words and two-word combinations
- **Stop words:** English stop words were removed

These settings help convert movie details into a numerical feature representation for the prediction model.

5.3 Importance of TF-IDF

Machine learning algorithms generally require numerical input rather than raw text. TF-IDF makes it possible to represent movie names, genres, directors, and actors numerically.

The resulting numerical features were passed to the Ridge Regression model through a Scikit-learn Pipeline. This pipeline combines text vectorization and model training into a single workflow.

By using TF-IDF, the project can learn patterns in the available movie descriptions and related information to estimate movie ratings. However, the predictions depend on the quality and usefulness of the features available in the dataset.

6. MODEL DEVELOPMENT AND TRAINING

6.1 Model Development

Model development is an important stage of a machine learning project. In this stage, the prepared dataset is used to build a model that can learn patterns and make predictions.

In this project, a machine learning pipeline was developed using Scikit-learn. The pipeline combines TF-IDF text vectorization and Ridge Regression. This makes it possible to transform movie-related text into numerical features and use those features to predict movie ratings.

6.2 Train-Test Split

The cleaned dataset was divided into training and testing sets using the `train_test_split()` function.

The dataset was split using the following settings:

- **Training data:** 80%
- **Testing data:** 20%
- **Random state:** 42

After splitting, the training set contained **6,335 samples**, while the testing set contained **1,584 samples**.

The training data was used to fit the model. The testing data was used to evaluate how well the trained model predicted ratings for data not used during training.

6.3 Ridge Regression

Ridge Regression is a supervised machine learning algorithm used for regression problems, where the output is a numerical value. It is based on linear regression and uses regularization to help control model complexity.

In this project, Ridge Regression was used to estimate movie ratings from the numerical features generated by TF-IDF.

The model was configured with an `alpha` value of `1.0`. This parameter controls the strength of regularization.

6.4 Machine Learning Pipeline

A Scikit-learn Pipeline was created to combine two main steps:

1. **TF-IDF Vectorizer:** Converts the combined movie text into numerical features.
2. **Ridge Regression:** Learns patterns from the training data and predicts movie ratings.

The pipeline was trained using the training features and their corresponding actual ratings. After training, it was used to generate predictions for the testing data.

6.5 Model Training Process

The model training process involved preparing the input features and target ratings, splitting the data, fitting the pipeline on the training set, and generating predictions for the test set.

This process allowed the project to develop a model for estimating movie ratings based on available movie information.

7. MODEL EVALUATION AND PERFORMANCE METRICS

7.1 Model Evaluation

Model evaluation is used to measure how accurately a machine learning model makes predictions. In this project, the trained Ridge Regression model was evaluated by comparing its predicted movie ratings with the actual ratings in the test dataset.

Three evaluation metrics were used: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 score.

7.2 Mean Absolute Error (MAE)

Mean Absolute Error measures the average absolute difference between actual and predicted values. A lower MAE indicates that the predictions are closer to the actual ratings on average.

The MAE obtained in this project was **0.875**. This means that the model's predictions differed from actual ratings by approximately 0.875 rating points on average in absolute terms.

7.3 Root Mean Squared Error (RMSE)

Root Mean Squared Error measures the square root of the average squared difference between actual and predicted values. It gives greater weight to larger prediction errors.

The RMSE obtained in this project was **1.124**. This metric helps evaluate the size of the model's prediction errors.

7.4 R^2 Score

The R^2 score indicates how well a regression model explains the variation in the target variable compared with a baseline that predicts the mean rating.

The model achieved an R^2 score of **0.320**. This indicates that the model captures some variation in movie ratings, although its predictive performance has room for improvement.

7.5 Summary of Results

The evaluation results obtained from the test dataset are shown below:

Evaluation Metric	Result
Mean Absolute Error (MAE)	0.875
Root Mean Squared Error (RMSE)	1.124
R^2 Score	0.320

7.6 Interpretation

The evaluation results show that the model can estimate movie ratings using the selected movie-related features. However, the R^2 score indicates that these features do not explain all the variation in actual ratings.

Movie ratings may also depend on factors not included in this model, such as audience preferences, reviews, and other contextual information. Therefore, the predictions should be treated as estimates rather than exact ratings.

8. PREDICTION RESULTS AND VISUALIZATION

8.1 Movie Rating Prediction

After training and evaluating the model, it was used to predict a rating for a sample movie. The sample movie details were provided as input to the trained machine learning pipeline.

The sample input included the following information:

- **Movie Name:** Example Movie
- **Genre:** Action Drama
- **Director:** Example Director
- **Actors:** Example Actor and Another Actor
- **Year:** 2026
- **Duration:** 120 minutes

For this sample input, the model predicted a movie rating of approximately **5.77**.

This is an example prediction generated from sample input data. It does not represent the actual rating of a real movie.

8.2 Actual and Predicted Ratings

The model generated predictions for the test dataset. These predicted ratings were compared with the corresponding actual ratings to evaluate how closely the model's estimates matched the real values in the dataset.

The notebook also displayed the average actual rating and average predicted rating. Both averages were approximately **5.84**.

Although these averages were similar, this does not mean that every individual prediction was accurate. The MAE, RMSE, and R^2 score provide additional information about the model's performance.

8.3 Visualization of Results

A scatter plot was created using Matplotlib to compare actual movie ratings with predicted movie ratings.

In this graph:

- The horizontal axis represents actual movie ratings.
- The vertical axis represents predicted movie ratings.

- A reference line represents the ideal situation in which predicted ratings equal actual ratings.

Points closer to the reference line indicate predictions closer to the actual ratings. Points farther from the line represent larger prediction errors.

8.4 Importance of Visualization

Visualization makes model performance easier to understand. It helps identify how closely the predicted ratings follow the actual ratings and whether prediction errors occur across different rating values.

The graph complements the numerical evaluation metrics and provides a visual understanding of the Ridge Regression model's prediction performance.

9. MODEL SAVING AND IMPLEMENTATION

9.1 Saving the Trained Model

After training and evaluating the model, the trained machine learning pipeline was saved for future use. Saving a model helps avoid retraining it every time a prediction is required.

In this project, the `joblib` library was used to save the trained pipeline in a file named `movie_rating_model.pkl`.

The saved pipeline contains the TF-IDF vectorizer and the Ridge Regression model. This allows the text-processing and prediction steps to be reused together.

9.2 Importance of Model Saving

Saving the trained model is useful for the following reasons:

1. It allows the trained pipeline to be reused later.
2. It reduces the need to train the model again for every prediction session.
3. It makes the model easier to integrate into a future application.
4. It helps preserve the trained model for further testing and development.

9.3 Using the Model for Prediction

The saved pipeline can be loaded later using `joblib`. New movie information must be prepared in the same feature format used during training before it is passed to the pipeline.

The pipeline then applies TF-IDF vectorization to the input text and uses Ridge Regression to generate a predicted rating.

The prediction is an estimate based on patterns learned from the training data. It should not be treated as a guaranteed or official movie rating.

9.4 Future Implementation

The saved model provides a foundation for developing a simple movie rating prediction application. A future version could include an interface where a user enters movie details and receives an estimated rating.

Such an application would require additional development and testing. The current project focuses on dataset preprocessing, model training, evaluation, prediction, and saving the trained pipeline.

10. CHALLENGES AND LIMITATIONS

10.1 Challenges Faced During the Project

During the development of the Movie Rating Prediction project, several important steps required attention and careful implementation.

1. Data Cleaning: The original dataset contained missing or invalid values. These needed to be handled before training the machine learning model.

2. Feature Preparation: Movie information was available in different columns, such as genre, director, actors, name, year, and duration. These details had to be combined into a single text feature for the model.

3. Text Vectorization: The textual movie information could not be directly used by the regression algorithm. TF-IDF was therefore used to convert text into numerical features.

4. Model Evaluation: The model's predictions needed to be compared with actual ratings using suitable evaluation metrics.

10.2 Limitations of the Project

The project has some limitations that should be considered when interpreting its results.

1. The model depends on the quality and completeness of the available dataset.
2. Only selected movie-related features were used for prediction.
3. The model does not directly consider audience reviews, viewer preferences, or other external factors.
4. The R^2 score of 0.320 indicates that the model explains only part of the variation in actual movie ratings.
5. Predicted ratings are estimates and may differ from actual or official ratings.

10.3 Possible Improvements

The project could be improved in the future by exploring additional useful features, testing different regression algorithms, and tuning model parameters. More extensive evaluation and better feature engineering could also help improve prediction performance.

These challenges and limitations provide useful learning opportunities for further development of the Movie Rating Prediction system.

11. CONCLUSION AND FUTURE SCOPE

11.1 Conclusion

The Movie Rating Prediction project was developed to apply machine learning techniques to the task of estimating movie ratings using movie-related information.

The project involved loading the IMDb Movies India dataset, cleaning the data, preparing relevant features, and converting textual information into numerical features using TF-IDF vectorization. A Ridge Regression model was trained using the prepared dataset and evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 score.

The model achieved an MAE of 0.875, an RMSE of 1.124, and an R^2 score of 0.320 on the test dataset. These results show that the model can learn some patterns from the selected features, although its prediction performance could be improved.

The trained pipeline was also saved using Joblib so that it could be reused in future work. Overall, this project provided practical experience in Python programming, data preprocessing, text feature extraction, machine learning model training, prediction, and evaluation.

11.2 Future Scope

The project can be extended in several ways:

1. **Feature Enhancement:** Additional relevant movie information could be explored to improve predictions.
2. **Algorithm Comparison:** Other regression algorithms could be tested and compared with Ridge Regression.
3. **Hyperparameter Tuning:** Model settings could be adjusted systematically to evaluate whether performance improves.
4. **Application Development:** A simple user interface could be created to allow users to enter movie details and receive predicted ratings.
5. **Further Evaluation:** Testing on additional data could help assess how well the model performs on different movies.

In conclusion, this project provides a foundation for further experimentation with machine learning-based movie rating prediction. Future improvements may help develop a more reliable and useful prediction system.

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12. REFERENCES

The following resources and documentation were used as references for the development of the Movie Rating Prediction project.

1. **IMDb Movies India Dataset**
   Dataset used for movie information and rating prediction in this project.
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   Python Software Foundation. Available at: https://docs.python.org/3/
3. **Pandas Documentation**
   Documentation for data loading, data manipulation, and data preprocessing. Available at: https://pandas.pydata.org/docs/
4. **NumPy Documentation**
   Documentation for numerical computing in Python. Available at: https://numpy.org/doc/
5. **Scikit-learn Documentation**
   Documentation for train-test splitting, TF-IDF vectorization, Ridge Regression, machine learning pipelines, and model evaluation metrics. Available at: https://scikit-learn.org/stable/
6. **Matplotlib Documentation**
   Documentation for creating data visualizations in Python. Available at: https://matplotlib.org/stable/
7. **Joblib Documentation**
   Documentation for saving and loading Python objects, including trained machine learning models. Available at: https://joblib.readthedocs.io/
8. **Google Colab**
   Cloud-based notebook environment used to write and execute the project code. Available at: https://colab.research.google.com/

These resources provide documentation and supporting information related to the tools, libraries, and techniques used in this project.